



Zoho Creator Assignment

24/05/2025

Application: Create a step-by-step application process for an Automated Costing System to be developed in ZOHO Creator.

By

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Key Features must have in Costing System:

Features	Sol	Click reference	Click Reference
(1) Variant Management	Using Forms.	Variant Management:	
(2) Cost Calculation	Using forms and workflows	Cost Calculation	
(3) Reporting	Using pages and charts	Reporting	
(4) Customization	Effective database design and few things included in the chart section	Conclusion	

Apps:

- ✓ Task 1: Create a step-by-step application process for an Automated Costing System to be developed in ZOHO Creator.*
- ✓ Task 2: Create a simple inventory management app where we can see live tracking of the fabrics & accessories inventories getting consumed in every order. How would you approach designing the database structure, forms & reports for this app?*

Variant Management:

The system allows the user to define and manage different product variants, like Premium, Value, Eco. For each variant, the system should store the respective price per square foot.

Database Structure/Form Structure:

Fabrics

Fabrics

All Fabrics

Fabric_details

We need **two** databases to store fabrics. So that we can maintain referential foreign key integrity.

Fabrics & Fabric_details **has one to many relationship.**

Fabrics table with unique fabric codes.

All Fabrics

	fabric_code	fabric name
	B70003	Georgette
	B70002	Chenille
	B70001	Chiffon

SORTING

price

	Fabrics	Fabric_code	Variant	price
	Satin	B70004	premium	250
	Chenille	B70002	value	85
	Chenille	B70002	eco	80
	Chiffon	B70001	premium	70
	Chiffon	B70001	value	65
	Satin	B70004	eco	60
	Georgette	B70003	eco	60
	Chiffon	B70001	eco	60
	Rubia	B70005	eco	40

Fabric details with **different price point** based upon the fabric code.

Cost Calculation

The system should calculate the total cost of each order based on the selected variant and the fabric consumption provided. The cost calculation should be derived by multiplying the fabric consumption by the respective price per square foot for the chosen variant.

Steps:

1. Create a form Order_Purchases.
2. Define fields using drag and drop feature.

☐	Client ID ▾	Fabrics ▾	variant ▾	Length ▾	Amount ▾	Order date ▾	Status ▾	Next Actions ▾	Fabric co... ▾
	Amy parker	Satin	eco	5	300	24-May-2025	Delivered	Update Customer	B70004
	Saify Kirmani	Satin	premium	10	2500	29-May-2025	Delivered	Request Feedback	B70004
	Yusuf Syed	Rubia	eco	10	400	31-May-2025	Processing	Confirm Inventory	B70005
	Lori green	Georgette	eco	70	4200	29-May-2025	In Production	Finalize Shipping Details	B70003
	Sarah speller	Chiffon	value	20	1300	24-May-2025	Shipped	Request Feedback	B70001
	Stacy Cooper	Chenille	eco	50	4000	27-May-2025	Processing	Update Customer	B70002
	David Simson	Chiffon	eco	170	13600	26-May-2025	In Production	Update Customer	B70001
	Kazim Syed	Satin	premium	5	1250	27-May-2025	Shipped	Update Customer	B70004

Orders_Purchases

Form Structure.

Client ID	Text	Lookup
Fabric	Number(Internal) because FK	Lookup → Fabrics
Variant	Number(Internal) because FK	Lookup → Fabric_details
Length	Number	5 (Default value: business rules)
Amount	Number	
Order date	date	
Status	Text	
Next Actions	Text	
Order time	Time	
Sys date	Time(System generated)	
Fab_code	Unique code	

Fabrics:

All Fabrics *

SORTING fabric_code ▲ ×

		fabric_code ▾	fabric name ▾	Image ▾	Url ▾
		B70001	Chiffon		chiffon
		B70002	Chenille		Chenille
		B70003	Georgette		Georgette
		B70004	Satin		Satin
		B70005	Rubia		Rubia

Fabrics_details

Fabric_details Report *

SORTING price ▼ ×

		Fabric_code ▾	Variant ▾	price ▾	Fabrics ▾
		B70004	premium	250	Satin
		B70002	value	85	Chenille
		B70002	eco	80	Chenille
		B70001	premium	70	Chiffon
		B70001	value	65	Chiffon
		B70004	eco	60	Satin
		B70003	eco	60	Georgette
		B70001	eco	60	Chiffon
		B70005	eco	40	Rubia

3. Write a workflow that updates the Amount field whenever there is a creation of new record or edit on {Fabric, variant, length} fields of the form.

Crux of the Cost Calculation → Services

User Action Flow:

Order_Purchases

	Select
Fabrics	-Select-
variant	-Select-
Length *	5
Order date *	25-May-2025
sys_date	25-May-2025
sys_time	13:07:08
Status	-Select-
Amount *	##### Func for Price of Fabric (variant) * length

Fabric code

variant

Length *

Order date *

Fabric code is a lookup field populated by Fabrics form.

(1)

Order_Purchases



Only available for
(1). eco
(2). value

System pops possible variant available for that particular fabric.

(2)

Fabric code

variant

Length *

Order date *

sys_date

sys_date

sys_time

End-user selects the variant of their choice.

Fabric code

variant

Length *

Order date *

sys_date

sys_time

Status

Amount *

As the user select or edits the value of the fields involved {fabric, variants, length}. The changes in Amounts are reflected instantaneously using a workflow.

Database structure for our Services

LTM_Services

☐	☐	Service ▾	price ▾
		welding	10
		cutting	5
		design	10

Client_Services

Client_Services Report

+

...

	Client ▾	Services ▾	Length ▾	Amount ▾	Ord_date ▾	Sys_Date ▾	Time ▾
	Kazim Syed	design	5	50	27-May-2025	25-May-2025	11:20:38
	Conan Obrien	welding	30	300	28-May-2025	25-May-2025	11:20:21
	Mark Cuban	design	40	400	27-May-2025	25-May-2025	11:19:58
	Lori Green	design	10	100	28-May-2025	25-May-2025	11:19:40
	Yusuf Syed	welding	25	250	29-May-2025	25-May-2025	11:13:24
	Kazim Syed	cutting	5	25	29-May-2025	25-May-2025	11:09:40

Cost Calculation using workflows.

TestApp

Order&Purchases

Home

Fabrics

Fabric_details

LTM_Services

Client_Services

Client_Services

Client_Services Report

Clients

Client_Services

Client *

Yusuf Syed

×

▽

Services *

welding

×

▽

Length *

25

↺

Amount

150

Ord_date *

29-May-2025

Sys_Date *

25-May-2025

Time *

11:13:24

Submit

Reset

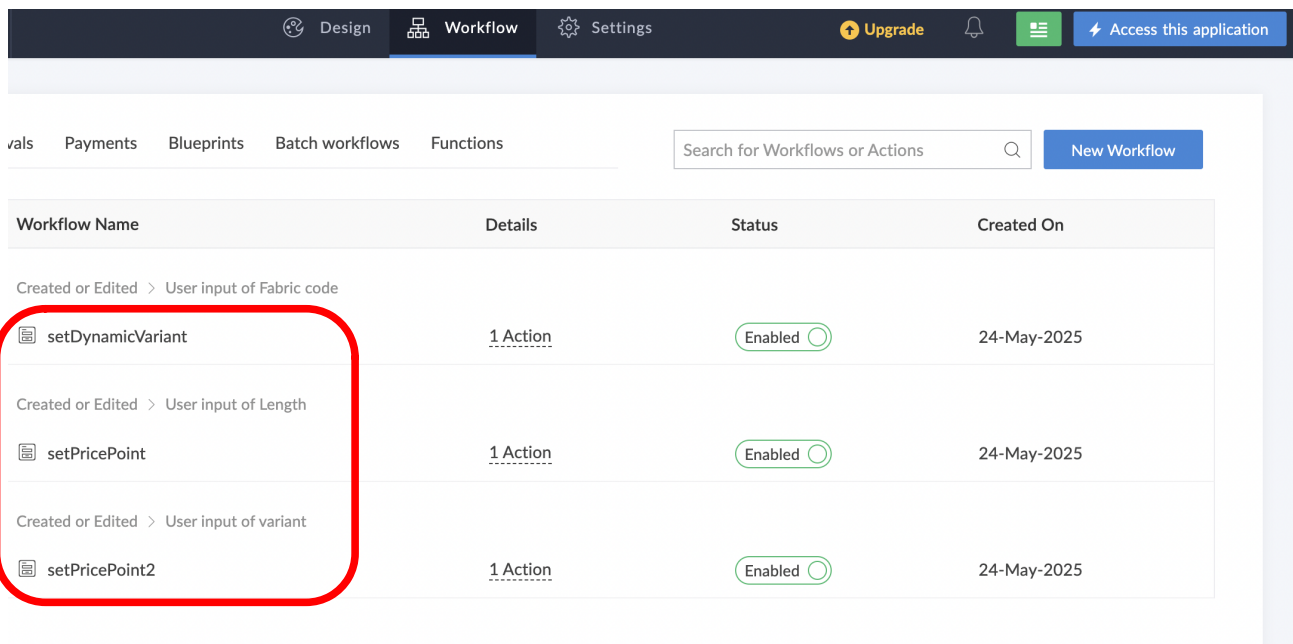
Here, we can see at length field. A small buffering indicating workflow is being used.
Here, 150 is the stale value.

Yusuf Syed	welding	25	250
------------	---------	----	-----

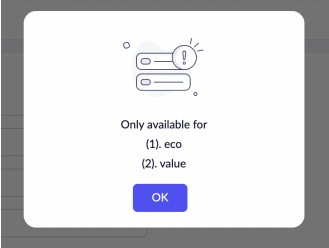
Updated value

Step 3 is to write workflows:

Workflow:



There are Three workflows.

Workflow	Executes when	Field	Purpose:
setDynamicVariant	(Created or edit) on Orders_Purchases	Fabric Code	 Tells all the variants available for this particular fabric
SetPricePoint	(Created or edit) on Orders_Purchases	Fabric_length	Instantaneous reflection to the calc.
setPricePoint2	(Created or edit) on Orders_Purchases	Fabric variant	Instantaneous reflection to the calc.

Reporting

The system should provide reporting capabilities to generate cost summaries for different orders. This can include displaying the total cost for each order, as well as aggregating costs by variant, order date, or other relevant criteria.

Steps:

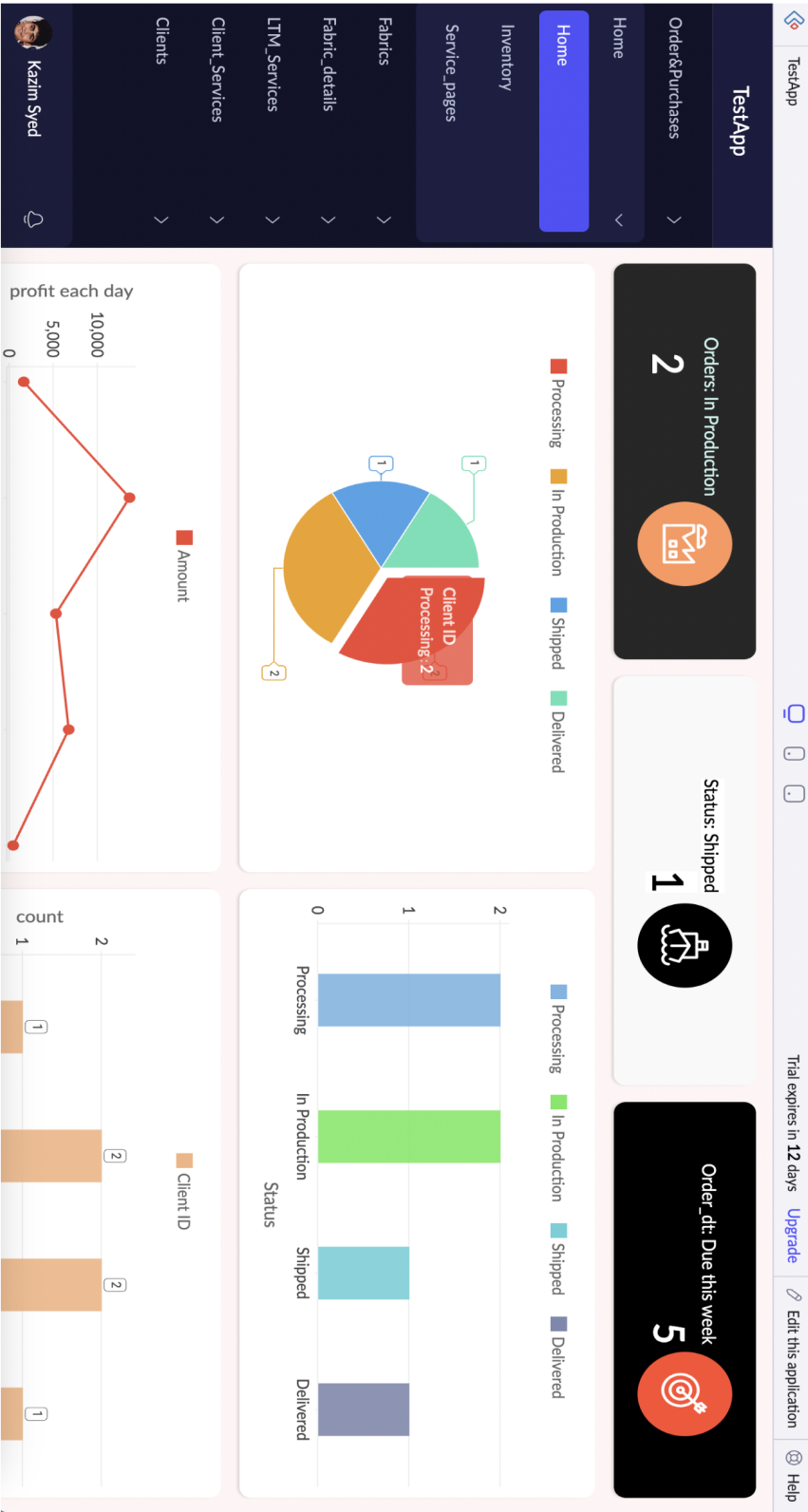
Step 1: Create a page.

Step 2: Using the drag and drop functionality, drop Panel and charts.

We have 3 Pages:

1. Home → Home
2. Home → Inventory
3. Home → Service_pages

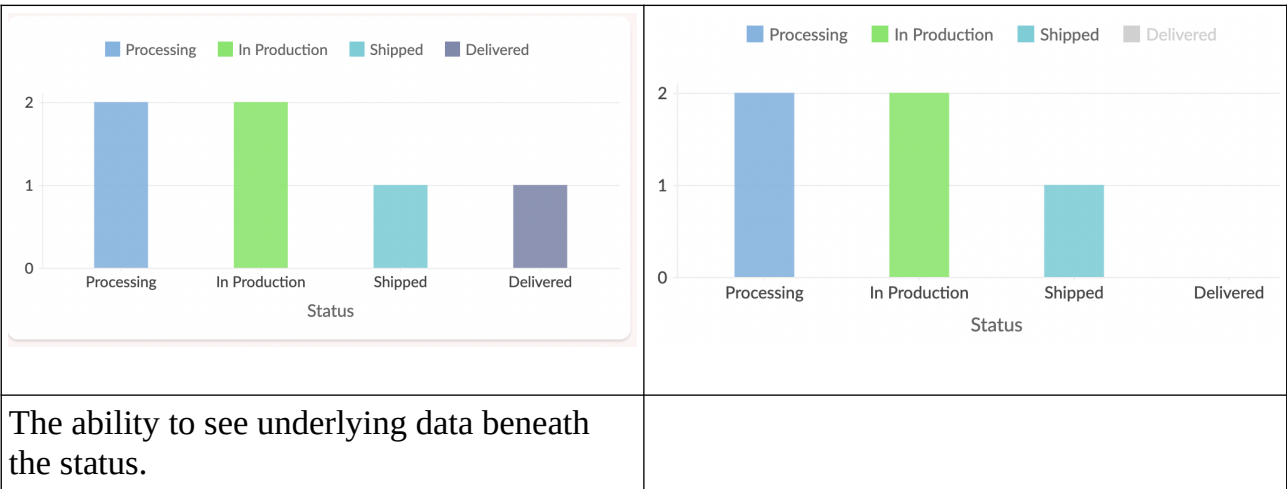
TestApp Order&Purchases Home Home Inventory Service_pages Fabrics Fabric_details LTM_Services Client_Services Clients	Note: I intend to change the names of this chart pages in the near future. Home: is for managing the logistics of our products and tracking orders. <table><tr><td>Logistics</td><td>Controls</td></tr><tr><td>Confirmed</td><td>When the order placed by our clients are ack by the system</td></tr><tr><td>Processing</td><td>The orders is processed.</td></tr><tr><td>In production</td><td>Creation of our final assembled product.</td></tr><tr><td>Waiting for shipping details</td><td></td></tr><tr><td>Shipped</td><td></td></tr><tr><td>Delivered</td><td>Product received by customer.</td></tr></table>	Logistics	Controls	Confirmed	When the order placed by our clients are ack by the system	Processing	The orders is processed.	In production	Creation of our final assembled product.	Waiting for shipping details		Shipped		Delivered	Product received by customer.
Logistics	Controls														
Confirmed	When the order placed by our clients are ack by the system														
Processing	The orders is processed.														
In production	Creation of our final assembled product.														
Waiting for shipping details															
Shipped															
Delivered	Product received by customer.														





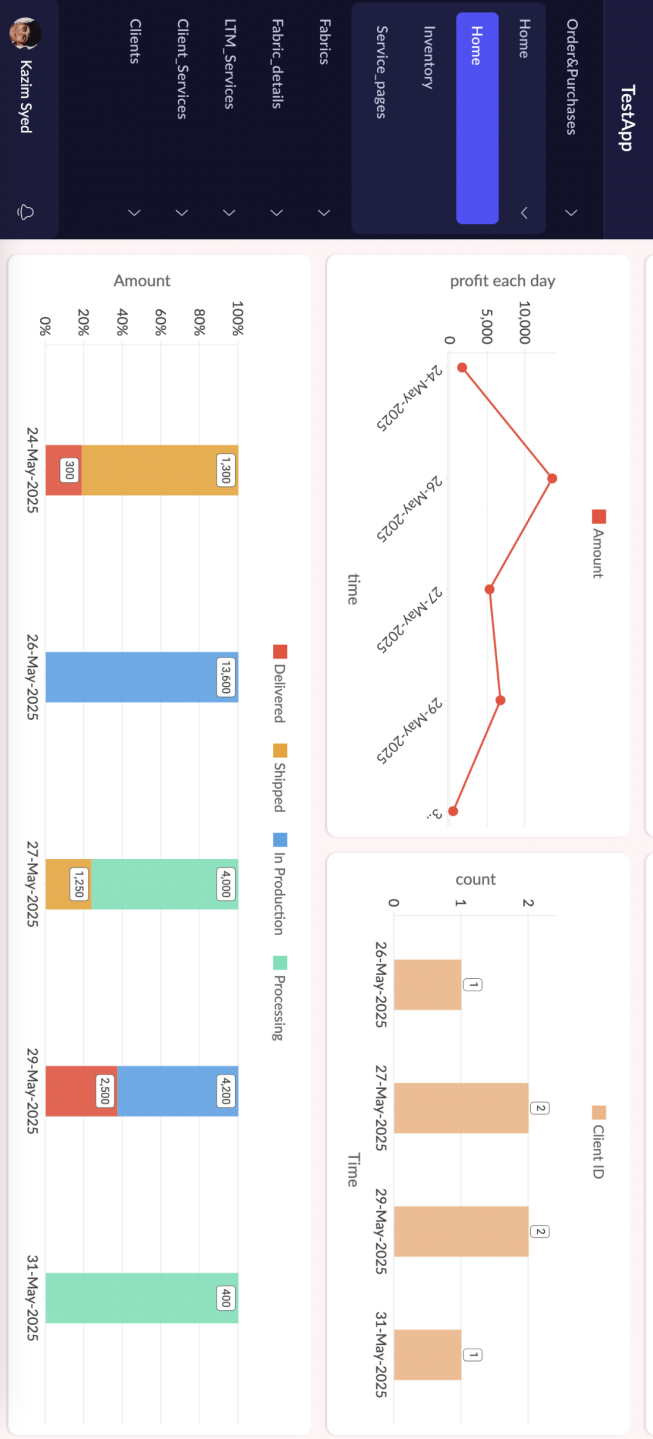
One of our enthused customization in the charts is belowed.

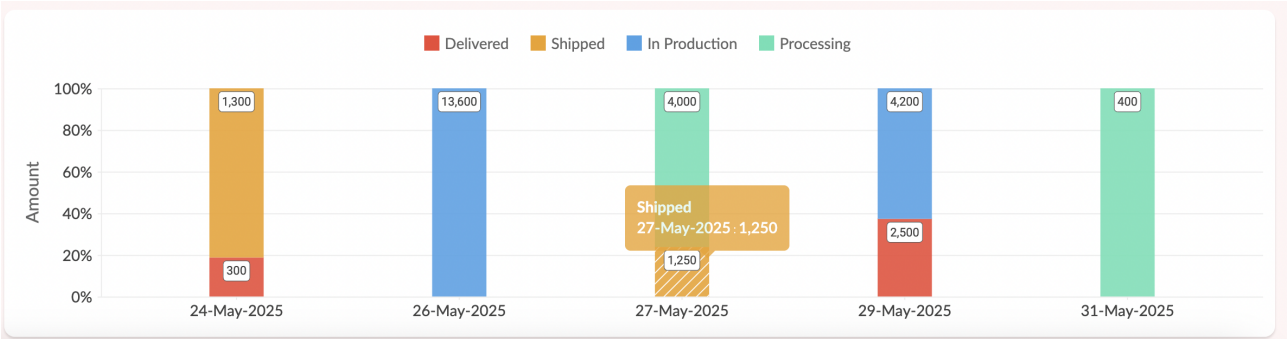
Once the order is delivered. Our attention can be allocated to the next fulfillment of product.



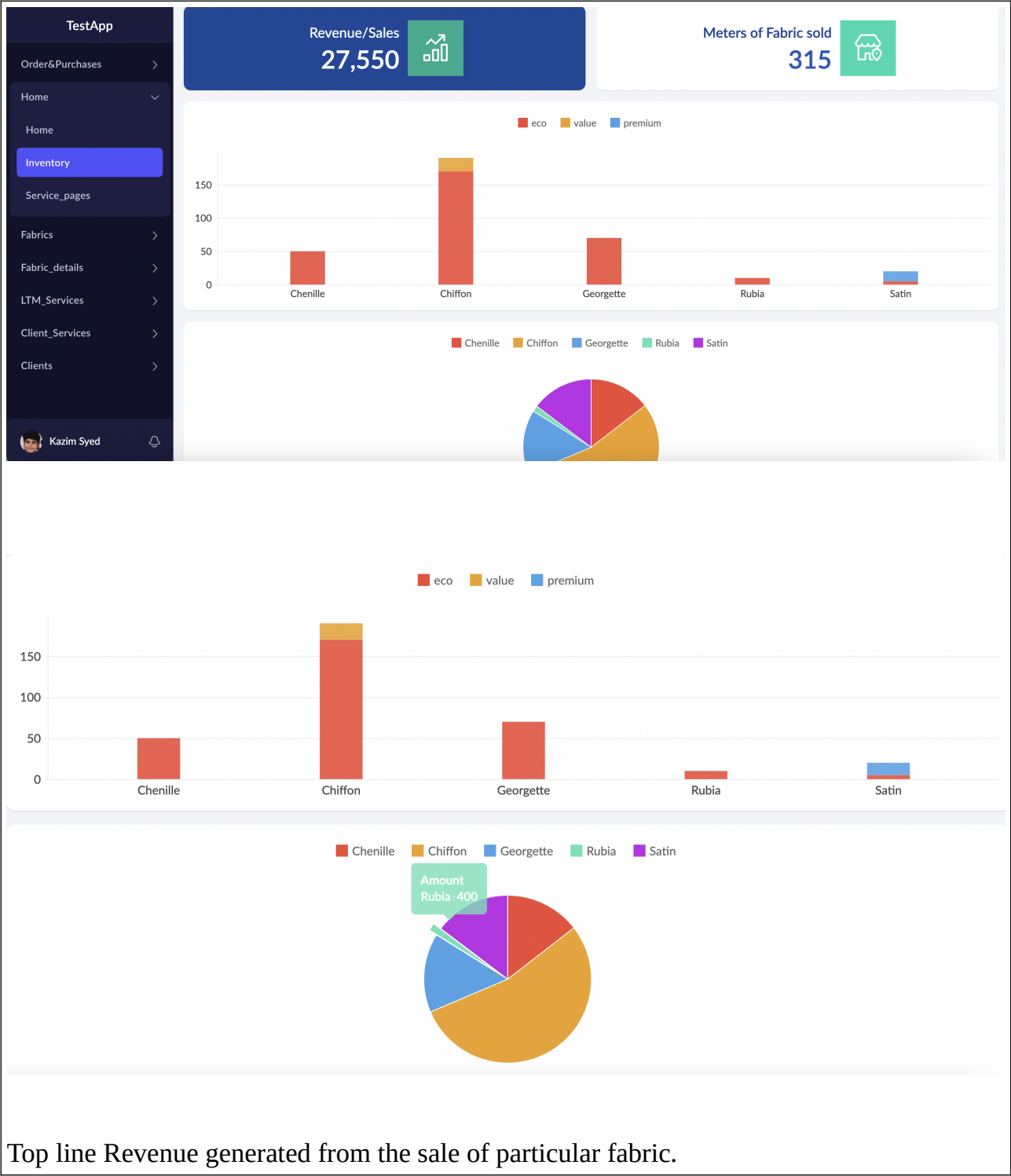
The ability to see underlying data beneath the status.

Order&Purchases Report							
Client ID	Fabrics	variant	Length	Amount	Order da...	Status	Next Actions
Yusuf Syed	Rubia	eco	10	400	31-May-2025	Processing	Confirm Inventory
Stacy Cooper	Chenille	eco	50	4000	27-May-2025	Processing	Update Customer





Home → Inventory



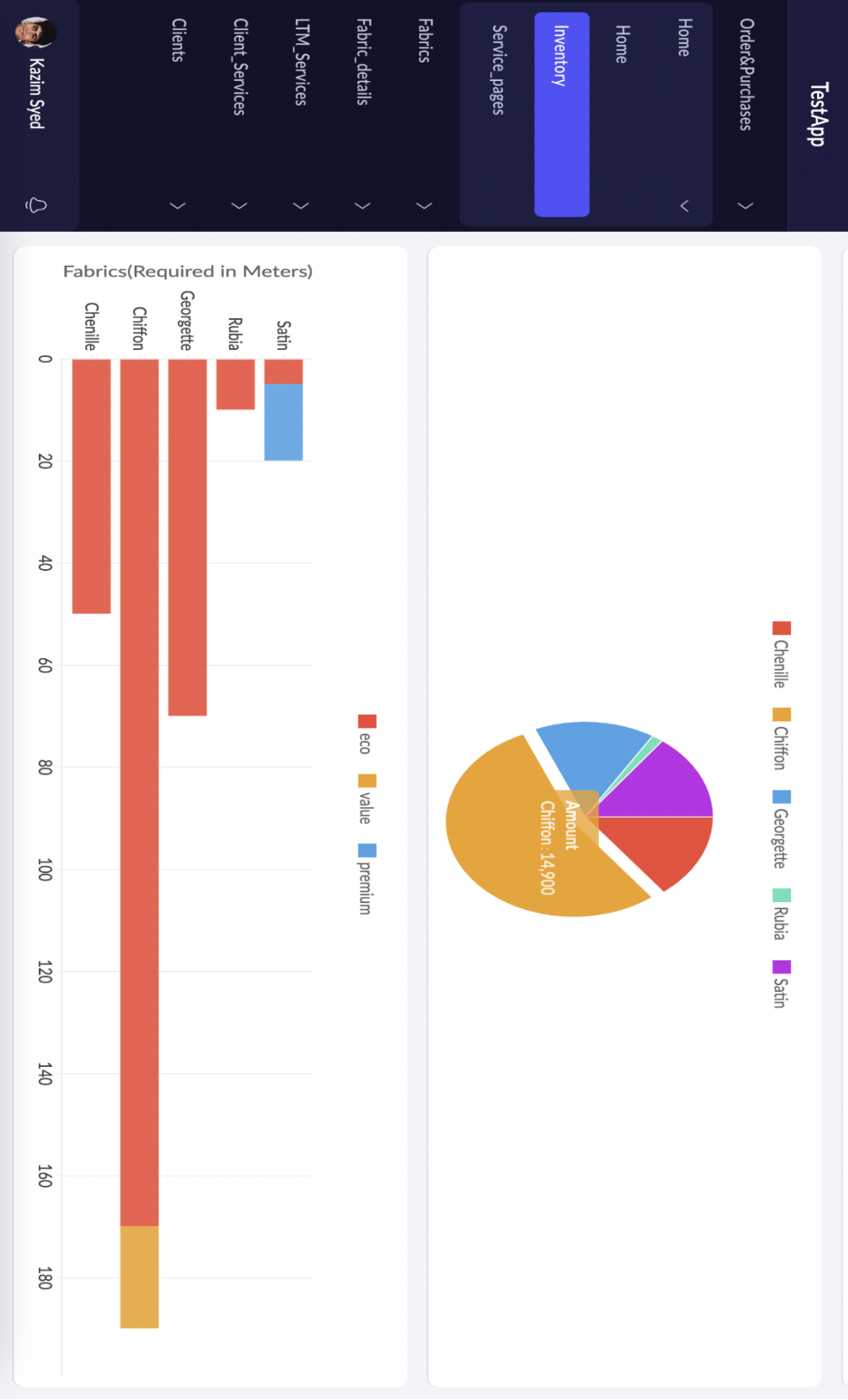
eco value premium

Fabric	eco	value	premium
Chenille	50	0	0
Chiffon	170	10	0
Georgette	70	0	0
Rubia	10	0	0
Satin	0	0	15

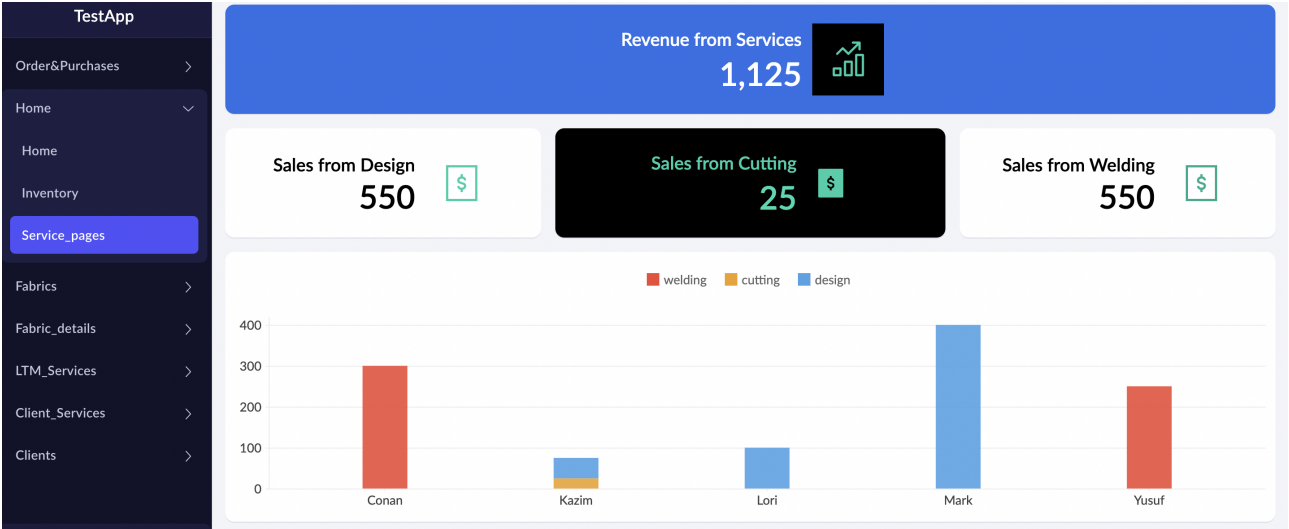
Chenille Chiffon Georgette Rubia Satin

Fabric	Amount
Chiffon	180
Chenille	50
Georgette	70
Rubia	10
Satin	15

Top line Revenue generated from the sale of particular fabric.



Home → Service Pages



Top line revenue generated by providing our clients with our services.

All Clients

☒	☐	client_nm ▾
☐	☐	Conan Obrien
☐	☐	Mark Cuban
☐	☐	Lori Green
☐	☐	Yusuf Syed
☐	☐	Kazim Syed

Customization

The system should allow for easy customization of variant names, prices, and other relevant parameters, as well as the ability to add or remove variants as needed.

Solution offered:

Effective database design is central to achieving ease of customization. Specifically, a database normalized to **Fourth Normal Form (4NF)** mitigates common issues such as data redundancy, along with insertion, update, and deletion anomalies.

...	fabric_...	fabric n...	Image	Url
...	B70001	Chiffon		chiffon
	B70002	Chenille		Chenille
	B70003	Georgette		Georgette
	B70004	Satin		Satin
	B70005	Rubia		Rubia



The url is mapped to the respected hyperlink image. Most of the other customization are shown in the reporting section of the document.

And as we can see we can very easily edit and update information in the database.

All Fabrics *		Trial expires in 12 days Upgrade Edit this application Help	
SORTING fabric_code ▲ ×		Edit Duplicate More ×	
fabric_code	fabric name	fabric_code B70001	
B70001	Chiffon	fabric name Chiffon	
B70002	Chenille	Image	
B70003	Georgette	Url chiffon	
B70004	Satin	Add a comment	
B70005	Rubia		

Fabric_details Report

price

	Fabric_code	Variant
...	B70004	premium
	B70002	value
	B70002	eco
	B70001	premium
	B70001	value
	B70004	eco
	B70003	eco
	B70001	eco
	B70005	eco

Edit

Duplicate

More

Fabric_code	B70004
Variant	premium
price	250
Fabrics	Satin

Add a comment

Note: I do believe this customization can be done better using a view of some sort. So even laymen can work through his way in the application. This is a MVP(minimum viable product from my end). I would be very enthused to bring better functionality and features in the final application.

Perks:

Gmail

Compose

Inbox7,068

- Starred
- Snoozed
- Sent
- Drafts98
- More

Labels+

- college attache...81
- Follow up115
- Job Status85
- My Own mails24
- My Reminders157
- Notes
- Read Through
- ..
- Upgrade→

Search mail

←📅⌚🗑️✉️📁⋮

Away?⚙️★☰👤

4 of 8,944<>

Hello,

A new entry added in 'Order_Purchases' form of 'TestApp' application.

User submitted data

Section	
Client ID	Amy parker
Fabric code	B70004
Fabrics	Satin
variant	eco
Length	5
Order date	24-May-2025
sys_date	24-May-2025
sys_time	22:32:22
Status	In Production
Amount	300
Order_time	22:32:00
Next Actions	Update Customer

Whenever a new orders is placed, its reported in an email to the Manager.

Workflow scripts

(1)

Workflow Scripts → setDynamicVariant

Deluge Script

[Order_Purchases](#) > Created or Edited > User input of Fabric code > setDynamicVariant

Deluge

Condition

if

else if

else

conditional if

Data Access

add record

fetch records

aggregate records

update record

```
1 variants_list = List();
2 fabric_records = Fabric_details[fab_code.fab_code == input.fab_code.fab_code] sort by price ASC;
3 // alert fabric_records;
4 op = "";
5 count = 1;
6 for each record in fabric_records
7 {
8     variants_list.add(record.variant);
9     // change `variant_name` to the actual field
10    // alert record.variant;
11    op = op + "\n(" + count + "). " + record.variant;
12    count = count + 1;
13 }
14 alert "Only available for" + op;
15 // variant.<field>:ui.add(<expression>);
16 // // // Now set the dropdown field choices
17 // variants:append("Kazim");
18
```

Output:

Order_Purchases

Fabric code

B70002

variant

-Select-

Length *

5

Only available for
(1). eco
(2). value

OK

or

Order_Purchases

Client ID *

Fabric code

B70003

Only available for
(1). eco

OK

(2)

Workflow Scripts → setPricePoint

Deluge Script

[Order_Purchases](#) > [Created or Edited](#) > [User input of Length](#) > [setPricePoint](#)

```
1 |if(input.fab_code != null && input.variant != null)
2 |{
3 |    //fab_code.fab_code is for fab_code lookup field in Fabric_details that is referencing Fabcode in Fabrics
4 |    try
5 |    {
6 |        fabric_records = Fabric_details[fab_code.fab_code == input.fab_code.fab_code && variant == input.variant];
7 |        if(fabric_records != null)
8 |        {
9 |            input.amount = fabric_records.price;
10 |            input.amount = input.amount * input.length_field;
11 |        }
12 |        else
13 |        {
14 |            alert "Fabric for that variant isn't available";
15 |        }
16 |    }
17 |    catch (e)
18 |    {
19 |        alert "Fabric for that variant isn't available";
20 |        info e;
21 |    }
22 |}
23
```

Fabric code	B70002
variant	eco
Length *	5
Order date *	24-May-2025
sys_date	24-May-2025
sys_time	16:41:46
Status	-Select-
Amount *	400

Conclusion

I have successfully fulfilled all the 4 features required in the assessment. I personally included an extra feature of logistics and tracking orders so we can fulfill our SLA agreements and meets our clients expectations and needs.

Future improvements:

Issue 1: Technical difficulty with Variant field. (Solvable in future).

Issue 2: More customization.

Issue 3: Need to find the accessibility to fire the same workflow on different fields.

Issue 4: Generates automatic reports and delivered it to the Manager in timely manner.

~Kind Regards

Kazim Syed

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